



10.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 16



Which of the following propositions is tautology?

- A. $(p \vee q) \rightarrow q$
C. $p \vee (p \rightarrow q)$

- B. $p \vee (q \rightarrow p)$
D. Both (B) and (C)

nielit2017july-scientistb-cs mathematical-logic

Answer key

10.0.2 NIELIT 2016 MAR Scientist C - Section C: 65



In propositional logic, which of the following is equivalent to $p \rightarrow q$?

- A. $\sim p \rightarrow q$
C. $\sim p \vee \sim q$

- B. $\sim p \vee q$
D. $p \rightarrow \sim q$

nielit2016mar-scientistc discrete-mathematics mathematical-logic

Answer key

10.0.3 NIELIT 2016 MAR Scientist C - Section C: 25



Which of the following is FALSE?

Read $\wedge$ as AND, $\vee$ as OR, $\sim$ as NOT, $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication?

- A. $((x \rightarrow y) \wedge x) \rightarrow y$
C. $(x \rightarrow (x \vee y))$

- B. $((\sim x \rightarrow y) \wedge (\sim x \wedge \sim y)) \rightarrow x$
D. $((x \vee y) \leftrightarrow (\sim x \vee \sim y))$

nielit2016mar-scientistc discrete-mathematics mathematical-logic

Answer key

10.1

First Order Logic (3)

10.1.1 First Order Logic: NIELIT 2017 DEC Scientist B - Section B: 43



Which one is the correct translation of the following statement into mathematical logic?

“None of my friends are perfect.”

- A. $\neg \exists x(p(x) \wedge q(x))$
C. $\exists x(\neg p(x) \wedge \neg q(x))$

- B. $\exists x(\neg p(x) \wedge q(x))$
D. $\exists x(p(x) \wedge \neg q(x))$

nielit2017dec-scientistb discrete-mathematics mathematical-logic first-order-logic

Answer key

10.1.2 First Order Logic: NIELIT 2021 Dec Scientist B - Section B: 25



Let $P(x)$ be “ x is perfect”, $F(x)$ be “ x is your friend” and the domain be all people. The statement, “At least one of your friends is perfect” is :

- A. $\forall x (F(x) \rightarrow P(x))$
- B. $\forall x (F(x) \wedge P(x))$
- C. $\exists x (F(x) \wedge P(x))$
- D. $\exists x (F(x) \rightarrow P(x))$

nielit-2021-it-dec-scientistb first-order-logic logical-reasoning

Answer key

10.1.3 First Order Logic: NIELIT 2022 April Scientist B | Section B | Question: 67



The first order logic statement $((R \vee Q) \wedge (P \vee \sim Q))$ is equivalent to which of the following?

- A. $((R \vee \sim Q) \wedge (P \vee \sim Q) \wedge (R \vee P))$
- B. $((R \vee Q) \wedge (P \vee \sim Q) \wedge (R \vee P))$
- C. $((R \vee Q) \wedge (P \vee \sim Q) \wedge (R \vee \sim P))$
- D. $((R \vee Q) \wedge (P \vee \sim Q) \wedge (\sim R \vee P))$

nielit2022apr-scientistb first-order-logic propositional-logic logical-reasoning

Answer key

10.2

Propositional Logic (2)

10.2.1 Propositional Logic: NIELIT 2017 July Scientist B (IT) - Section B: 13



Which of the following statements is false?

- A. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $\sim Q \wedge \sim P$
- B. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee P$
- C. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee (P \wedge \sim Q)$
- D. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $P \vee (Q \wedge \sim P)$

nielit2017july-scientistb-it mathematical-logic propositional-logic

Answer key

10.2.2 Propositional Logic: NIELIT Scientist B 2020 November: 84



Given the truth table of a Binary Operation \$ as follows:

X	Y	$X \oplus Y$
1	0	1
1	1	0
0	1	1
0	0	0

Identify the matching Boolean Expression.

- A. $X \oplus \neg Y$
 C. $\neg X \oplus \neg Y$

- B. $\neg X \oplus Y$
 D. none of the options

nielit-scb-2020 mathematical-logic propositional-logic discrete-mathematics

Answer key 

Answer Keys

10.0.1	C
10.1.3	C

10.0.2	B
10.2.1	A

10.0.3	D
10.2.2	D

10.1.1	A
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10.1.2	C
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